

JavaScript Required for Java Full-Stack Development

Variables

Store data values. Use `let` for changeable values and `const` for constants.

Example: `let name = 'John'; const age = 25;`

Data Types

Represent different kinds of data such as strings, numbers, booleans, arrays, and objects.

Example: `let active = true; let score = 99;`

Operators

Perform calculations and comparisons. Used in conditions and expressions.

Example: `a + b; age >= 18;`

Conditionals

Run different code paths based on conditions.

Example: `if (age >= 18) { console.log('Adult'); }`

Loops

Repeat code multiple times until a condition is met.

Example: `for(let i=0;i<5;i++){ console.log(i); }`

Functions

Reusable blocks of code that perform specific tasks.

Example: `function greet(){ return 'Hi'; }`

Arrays

Store multiple values in a single variable.

Example: `const nums = [1,2,3];`

Array Methods

Transform and filter data efficiently.

Example: `nums.filter(n => n % 2 === 0);`

Objects

Store related data as key-value pairs.

Example: `const user = {name: 'John', age: 25};`

Strings

Work with text and perform text manipulation.

Example: `'hello'.toUpperCase();`

DOM Basics

Access and modify webpage elements.

Example: `document.querySelector('#btn');`

Events

Respond to user actions like clicks and typing.

Example: `button.addEventListener('click', fn);`

Arrow Functions

Shorter syntax for writing functions.

Example: `const add = (a, b) => a+b;`

Destructuring

Extract values from objects and arrays easily.

Example: `const {name} = user;`

Spread Operator

Copy or merge arrays and objects.

Example: `const arr2 = [...arr1];`

JSON

Format used for data exchange between frontend and backend.

Example: `JSON.stringify(user);`

Promises

Handle asynchronous operations.

Example: `fetch(url).then(res => res.json());`

Async/Await

Cleaner syntax for asynchronous code.

Example: `const data = await fetch(url);`

Fetch API

Send HTTP requests to backend APIs.

Example: `await fetch('/api/users');`

Error Handling

Catch and manage runtime errors.

Example: `try {} catch(err) {}`

Form Validation

Validate user input before submission.

Example: `/^[^\s@]+@[^\s@]+\.[^\s@]+$`